

Simple: There exists some cuisine which is liked by atleast one of the students Monique Arsenault ~~and~~ Jay Johnson.

d) $\forall x \forall z \exists y ((x \neq z) \rightarrow \neg (T(x, y) \wedge T(z, y)))$

From my school, for all students x and for all students z , there exists some cuisine y s.t. if x and z are not the same person then it is not the case that student x likes cuisine y and student z also likes cuisine y together.

Simple: For all pair of students from my school, there exists some cuisine which cannot be liked by both the students together.

e) $\exists x \exists z \forall y (T(x, y) \rightarrow T(z, y))$

From my school, there exists a student x and there also exists a student z such that for all cuisines y , x likes cuisine y if and only if z also likes cuisine y .

Simple: There exists a pair of students from my school who likes the exact same cuisines.

f) $\forall x \forall z \exists y (T(x, y) \leftrightarrow T(z, y))$

From my school, for all students x and for all students z , there exists a cuisine y such that x likes y if and only if z likes y .

Simple: For all pair of students from my school, there exists some cuisine which is liked together or disliked together by the pair of student.

g) Let $Q(x, y)$ be the statement "Student x has been a contestant on quiz show y ." Express each of these sentences in terms of $Q(x, y)$, quantifiers and logical connectives, where the domain for x consists of all students at your school and for y consists of all quiz shows on television.

a) There is a student at your school who has been a contestant on a television quiz show: $\exists x \exists y Q(x, y)$